Original Article

Magnetic Resonance Imaging Image-Based Segmentation of Brain Tumor Using the Modified Transfer Learning Method

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Abstract

Purpose: The goal of this study was to improve overall brain tumor segmentation (BraTS) accuracy. In this study, a form of convolutional neural network called three-dimensional (3D) U-Net was utilized to segment various tumor regions on brain 3D magnetic resonance imaging images using a transfer learning technique. **Materials and Methods:** The dataset used for this study was obtained from the multimodal BraTS challenge. The total number of studies was 2240, obtained from BraTS 2018, BraTS 2019, BraTS 2020, and BraTS 2021 challenges, and each study had five series: T1, contrast-enhanced-T1, Flair, T2, and segmented mask file (seg), all in Neuroimaging Informatics Technology Initiative (NIFTI) format. The proposed method employs a 3D U-Net that was trained separately on each of the four datasets by transferring weights across them. **Results:** The overall training accuracy, validation accuracy, mean dice coefficient, and mean intersection over union achieved were 99.35%, 98.93%, 0.9875%, and 0.8738%, respectively. **Conclusion:** The proposed method for tumor segmentation outperforms the existing method.

Keywords: Convolutional neural networks, deep learning, transfer learning, three-dimensional image processing

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Introduction

A radiologist[1] interprets the huge quantity of images produced each day for diagnosis through visual processing, which takes time and is highly susceptible to human error. As a result, automatic segmentation using artificial intelligence and deep learning is currently used to eliminate this inaccuracy. Segmentation[2] is a common method for medical image analysis, i.e., based on a region of comparable features including gray level, contrast, and pixel value. In the context of medical images, the major goal of segmentation was to

1. Separate different parts of the region of interest (ROI), i.e., tumors and normal structures
2. Investigate the anatomical structures
3. After any treatment, such as chemotherapy or radiation therapy, measure the volume of ROI.

However, segmentation using only one type of image is a difficult task since no single imaging modality can offer all of the information.[3] As a result, the above information from multimodality imaging can improve the segmentation accuracy and reliability.[4,5] Deep learning and artificial intelligence show great potential for the identification and segmentation of medical images using magnetic resonance imaging (MRI) because it simplifies the process of automating imaging-based diagnosis. Using the brain tumor segmentation (BraTS)[6] dataset, Zhou *et al*.[7] employed four different modalities (T1-weighted, T2-weighted, contrast-enhanced [CE] T1-weighted, and fluid attenuation inversion recovery [FLAIR]) of MRI images for multi-modal image segmentation. To achieve tumor segmentation, they deploy three convolutional neural networks (CNNs) in a cascade. DeepSeg, a completely automated CNN approach based on an ensemble of two encoder-decoder methods, was employed by Zeineldin *et al*.[8] To increase the detection of the entire tumor and its subregions in comparison to the total tumor, they used all accessible MRI modalities (T1, T2, T1ce, and T2-FLAIR), as well as

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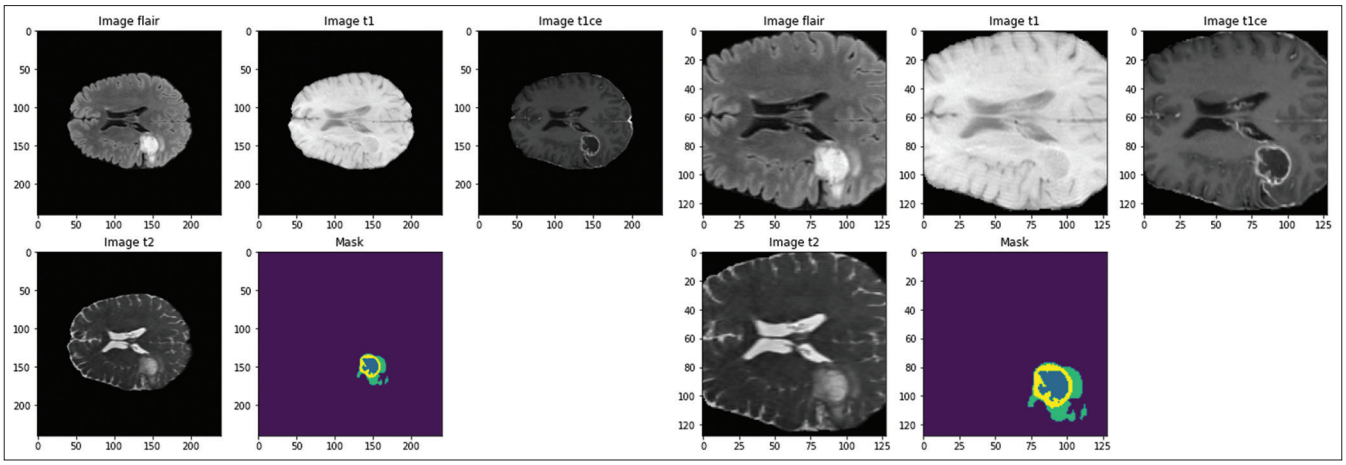


Figure 1: Left (Image before processing) and Right (Image after processing), size decreased to 128-by-128-by-128

region-based training and excessive data augmentation. Roy Choudhury *et al*.[9] used DeepLabv3 + as the base network for BraTS in multimodal MRI images in the BraTS dataset, with Xception[10] as the base network. They used T1-weighted, T2-weighted, CE-T1-weighted, and FLAIR images to segment the total tumor, tumor core, and enhancing tumor (ET), similar to Wang *et al*.[11] They employed two-dimensional orthogonal slices derived from the three-dimensional (3D) volume as input instead of the entire volume. In this study, using a publically available dataset from the Multi-Modal BraTS Challenge, we developed a new approach for multi-modal segmentation of brain tumors. We use various BraTS datasets to improve the neural network's accuracy by transferring the weight of the training process.

The lack of sufficient datasets and resources to train the model is a fundamental issue in the field of deep learning and machine learning. Due to this, first, the model's performance suffers as a result of the lack of data, and second, training a machine learning model takes either a long time or a high-end machine with plenty of graphics memory, among other things. As a result, transfer learning is quite beneficial in this situation. Transfer learning, also known as knowledge transfer,[12] is a strategy for reusing a previously learned model to address a new problem. This saves time by not having to retrain the model for similar tasks, and we can also take a dataset from another domain to train our model and reuse it for our purposes.

Materials and Methods

High-grade gliomas (HGG), low-grade gliomas (LGG), and other forms of brain cancers have been observed in humans.[13] These tumors come in a variety of shapes and sizes, and their characteristics change from country to country.[14] As a result, training a model just on a dataset from a single region can result in poor overall performance. The major goal of this study was to improve the overall performance of the image segmentation model on various datasets from various regions and for various types of tumors without relying on human assistance. Federated segmentation[15] was the name for this form of segmentation. A model that had been trained on a variety of datasets and tumor types can outperform a model that had only been trained on a single dataset. Tumors of the HGG and LGG kinds were found in our 2240-study database.

Data collection

A publicly available multi-institutional dataset, BraTS 2018,[7,16,17] BraTS 2019,[7,16,17] BraTS 2020,[7,16,17] and BraTS 2021[7,16,18] having 3D multi-modal images, was used for this task of segmenting brain tumors using. This dataset was released by the Center for Biomedical Image Computing and Analytics (CBICAs), University of Pennsylvania, and downloaded from the CBICA image processing portal. This dataset includes four types of MRI images (T1-weighted, T2-weighted, CE-T1-weighted, and FLAIR) and one segmented file, all in NIFTI format. Non-ET core (NETC), enhancing tumor core (ETC), and peritumoral edema were the different tumor-type sections on the segmented mask file. There were 2240 MRI scans in the dataset. We have used the weight carrier transfer learning method for segmentation and evaluated accuracy based on various parameters such as the Jaccard coefficient or intersection over union (IOU) score, dice score, and pixel accuracy. The code was run using Google Collaboratory Pro[19] and had Python V3.7. It offers 25 GB of RAM and a Tesla P100 GPU that had 16 GB of graphics memory. For medical image segmentation, there were numerous deep-learning network designs were available. We employed 3D U-Net[20] in our study because compared to other segmentation algorithms such as VGG16, VGG19, and inception 3D U-Net shows superior performance in qualitative metrics such as Jaccard coefficient, Dice coefficient, and pixel accuracy, and it was very useful for complex medical MRI image segmentation. For testing reasons, a total of 100 MRI scans were employed. For training and validation, 75% and 25% of the data were used, respectively, accounting for a total number of 1605 studies for training and 535 studies for validation, respectively.

Data processing

The dataset was 240-by-240-by-155 pixels in size, which was too much for the system to handle, so we processed it to remove slices with a lot of background pixels. As a result of processing, the image used as input and output had final dimensions of 128-by-128-by-128 as shown in Figure 1. First, we performed the segmentation task using all four modalities (T1-weighted, T2-weighted, CE-T1-weighted, and FLAIR). All the four modalities (T1-weighted, T2-weighted, CE-T1-weighted, and FLAIR) were stacked together along the z-axis, which makes the data 4 channels input. Therefore, the input layer of the 3D U-Net was changed to allow input and output for the appropriate number of channels, i.e., four (128-by-128-by-128-by-4). The architecture was designed in such a way that it accepts input in the channel last format (i.e., IMG_HEIGHT, IMG_WIDTH, IMG_DEPTH, IMG_CHANNELS). In the output layer of CNN, the activation function was changed from rectified linear unit (ReLU) to softMax, because ReLU was a type of activation function which will output the input directly if it was positive or one, otherwise it will give output as zero. Hence, we used ReLU in the Single-Label problems, where the output was either 1 or zero. For multiclass-label, softMax was used in the output layer of the neural network. SoftMax[21] is a mathematical function that converts a vector of numbers into a vector of probabilities, where the probabilities of each value were proportional to the relative scale of each value in the vector.

Modified transfer learning method

A bridge connects the encoder and decoder subnetworks of the original U-Net.[22] There were numerous stages (depth) in these networks, each with multiple layers. Each encoder step consists of two convolution paths with activation ReLU[23] followed by a 2-by-2 max-pooling layer. In a transposed fashion, the same was followed in the decoder path. The training optimizer Adaptive Moment Estimation (ADAM)[24] was utilized. We employed a loss function that combined dice loss[25] and focal loss, as well as a learning rate of 0.00003, to get good results after a few training cycles. To avoid memory difficulties, a batch size of 5 images was employed. The maximum number of epochs that were used was 100.

First, we created a 3D U-Net model from scratch, complete with encoder and decoder parts. The model had then trained on the BraTS 2018 dataset, which had been preprocessed to make it smaller and fit into system memory. The model architecture and weights can be saved in a single file using the TensorFlow library,[26] which was used in this work. This file can be used to create a new model with the same architecture and weights that have already been trained. As shown in Figure 2, the trained model, including its architecture and trained weights w^1_{ij} (i, was the number of neurons in the preceding layer or first layer, and j, was the number of neurons in the next layer of a CNN), was preserved. Now, these trained weights, w^1_{ij}, and architecture were used to create a new model that would be trained on the BraTS 2019 dataset, and again the model's weights, w^2_{ij}, were saved after training. For the next two datasets, BraTS 2020 and BraTS 2021, the process was repeated. After all four datasets were trained, the final model was saved and used to predict the test dataset. The strategy enables the model to learn more effectively on varied datasets while also assuming a prior knowledge of the job, resulting in improved accuracy and other segmentation performance indicators.

Results and Discussions

Metrics for evaluation

In our research, we used a variety of segmentation performance metrics,[27] as stated in the equations (1–4). The IOU score, known as the Jaccard coefficient, is the most used statistic for image segmentation model performance, shown in Figure 3. It is useful when we need statistically precise measurements with a penalty for false negatives (FNs). The percentage of correctly detected pixels for each class (i.e., NETC and ETC) is called accuracy. The weighted dice similarity coefficient (DSC) is the dice of each class weighted by the number of pixels in that class. When images have disproportionally sized classes, the weighted DSC is employed to lessen the influence of errors in the small classes on the overall quality score The DSC or f1 score is defined as twice the number of elements common to both sets (i.e., ground truth and predicted mask) divided by the sum of the number of elements in each set, shown in Figure 4. The fraction of pixels in an image that is correctly categorized is known as pixel accuracy.

The IOU score for sets A and B was calculated as follows:

$$IOU(A, B) = |Intersection(A, B)|/|Union(A, B)| \quad (1a)$$

where | A| represents the cardinal of set A. The IOU score can also be expressed in terms of true positives (TP), false positives (FP), and FN as:

$$IOU(A, B) = TP/(TP + FN + FP) \quad (1b)$$

where TP = True Positives, FP = False positives, FN = False Negatives

The DSC or f1 score for two given sets A and B was expressed as:

$$DSC = 2* | intersection(A, B)|/(|A| + |B|) \quad (2)$$

where | A| and | B| were the cardinalities of the two sets (i.e., the number of elements in each set).

Loss functions

In situations where there was a problem with Data-imbalancement[28] (background was also labeled and had a high portion as compared to ROI), loss functions were typically used. Because the region of interest in medical images was so small in comparison to the background, there was a high probability that the learning process will become stuck in local minima of loss and produce a model with predictions that were heavily biased in favor of the background. In this scenario, the area of interest or tumor will either be partially segmented or missed. Dice loss[25] was therefore used to keep a check on this issue. However, as noted by Zhang Y *et al*.,[29] there were

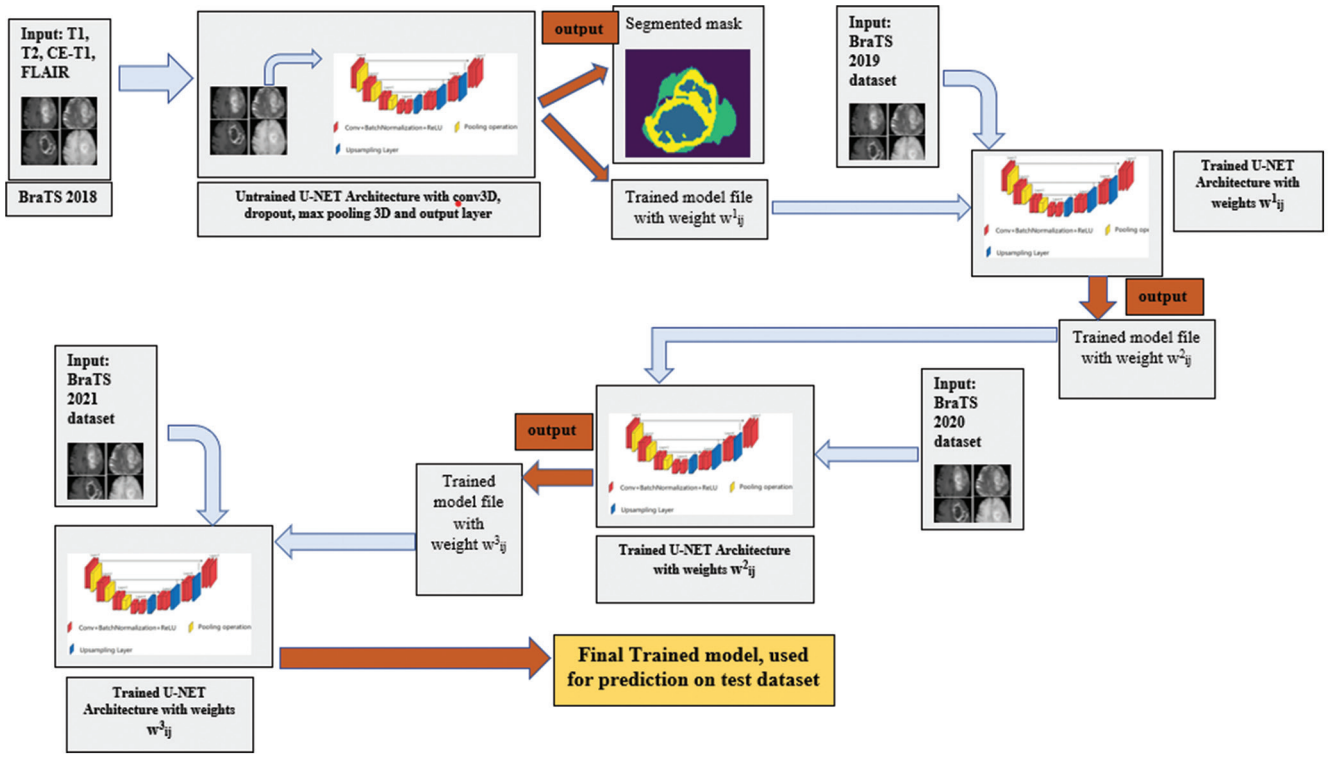


Figure 2: Proposed model with transfer learning method

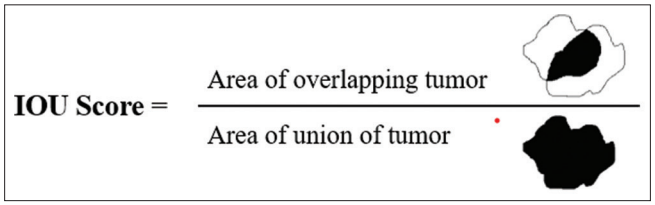


Figure 3: Illustration of IOU score: Area of overlap/area of union. IOU: Intersection over union

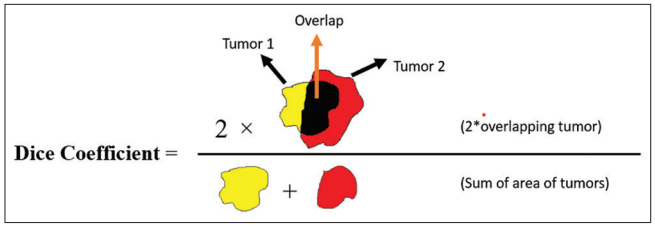


Figure 4: Illustration of Dice similarity coefficient: 2x overlap/Total number of pixels

a number of drawbacks to dice loss, including (a) that it was insensitive to the distance of nonoverlapping regions, (b) that it was insensitive to the precise contour of segmented regions because the regions only have a minor impact on the overall volume, and (c) that it had a large dynamic sensitivity to small lesions compared to large lesions. Therefore, we used two different loss functions,[29,30] the first of which was dice loss, and the second of which was a total loss,[29] which was a combination of dice loss and categorical focused loss. Figure 5 shows segmentation results on different test images with their corresponding ground truth. The dice loss was calculated as follows:

$$\text{Dice (loss)} = 1\text{-DSC} = 1\text{-}(2^* \mid \text{intersection (A, B)} \mid / (\mid A \mid + \mid B \mid)) \quad (3)$$

and the total loss was

$$\text{Total loss} = \text{dice (loss)} + (2^* \text{ Categorical Focal loss [CFL]}) \quad (4)$$

where CFL is expressed as:

$$\text{CFL (A, B)} = -A^* \alpha^* (1 - B)\, \gamma^* \log (B) \quad (5)$$

where A was ground truth, B was prediction or probability of ground truth class, α was float or integer (default value = 0.25), and γ was float or integer value (default value = 2.0).

Prediction of different test images

Table 1 shows the results of some of the test images using the model outlined in section 2.1. As can be seen, the model's accuracy was >99%, which was far higher than existing state-of-the-art frameworks. The Jaccard coefficient or Mean IOU score was also above 0.855, and the mean pixel accuracy was more or equivalent to that of the existing frameworks.

Comparison after training on every dataset

We also examined the performance of different segmentation parameters after training on the datasets from each year, as given in Table 2. The model's accuracy increased by approximately 4%, from 95.9% to 99.3%. The average IOU

Table 1: Segmentation results of the test dataset

Image number	Accuracy of model	Mean IOU	Mean dice score (for the WT)	Weighted dice score		Mean pixel accuracy
				NETC	ETC	
1	0.991	0.858	0.991	0.961	0.905	0.993
2	0.981	0.859	0.980	0.855	0.849	0.936
3	0.997	0.861	0.997	0.868	1.000	0.968
4	0.993	0.810	0.992	0.906	0.901	0.956
5	0.996	0.866	0.996	0.937	1.000	0.999
6	0.991	0.858	0.991	0.926	0.893	0.993
7	0.987	0.872	0.986	0.906	0.896	0.986
8	0.994	0.868	0.994	0.961	0.855	0.947
9	0.994	0.883	0.994	0.886	0.934	0.995
10	0.988	0.875	0.988	0.910	0.908	0.997

ETC: Enhancing Tumor core, NETC: Non-ETC, IOU: Intersection over union, WT: Whole tumor

Table 2: Comparison of segmentation parameters after every training step

Brats' dataset year	Accuracy of model	Mean IOU	Mean dice score (for the WT)	Weighted dice score		Mean pixel accuracy
				NETC	ETC	
2018	0.959	0.711	0.969	0.805	0.766	0.960
2019	0.979	0.826	0.979	0.891	0.880	0.961
2020	0.989	0.849	0.989	0.894	0.875	0.988
2021	0.993	0.880	0.992	0.917	0.914	0.991

ETC: Enhancing Tumor core, NETC: Non-ETC, IOU: Intersection over union, WT: Whole tumor

Table 3: Comparison of segmentation parameters with different loss functions

Loss function	Accuracy of model	Mean IOU	Mean dice (for WT)	Weighted dice score		Mean pixel accuracy
				NETC	ETC	
Dice loss	0.993	0.880	0.992	0.917	0.914	0.998
Total loss	0.991	0.850	0.991	0.901	0.883	0.989

ETC: Enhancing Tumor core, NETC: Non-ETC, IOU: Intersection over union, WT: Whole tumor

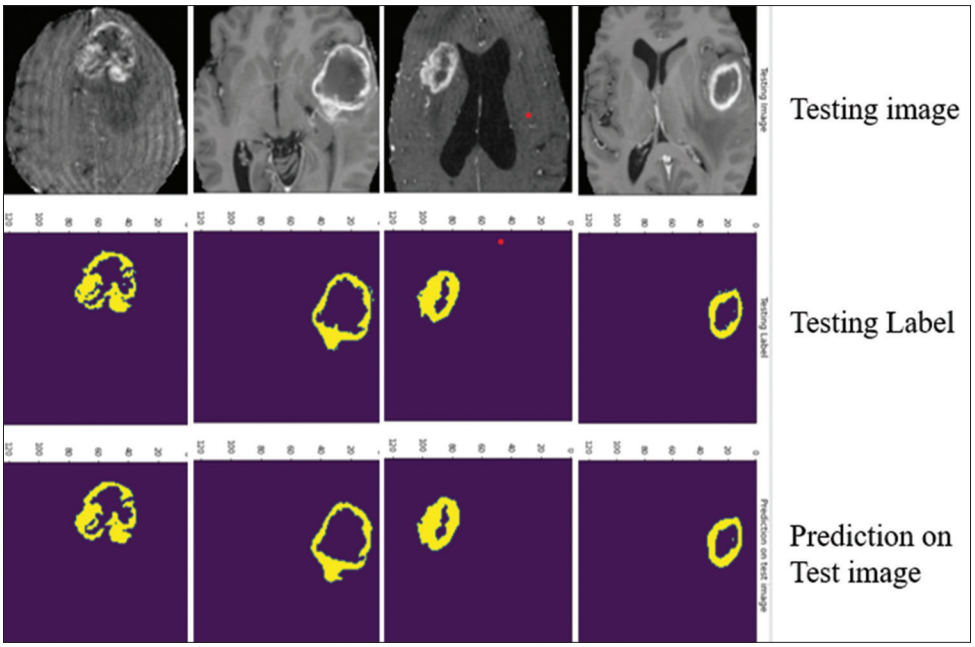


Figure 5: Segmentation results on different test images, showing (from up to down) Input image, original segmented tumor or ground truth, and predicted tumor or segmented tumor

Table 4: Comparison of the proposed method with existing methods

Technique	Accuracy	Mean IOU	Mean dice (for WT)
Zhou *et al*.[7]	0.9753±0.008	0.8305±0.007	0.9684±0.008
Roy Choudhury *et al*.[9]	0.9810±0.008	0.8375±0.006	0.9650±0.007
Wang *et al*.[11]	0.9825±0.008	0.8384±0.006	0.9688±0.007
Arif *et al*.[31]	0.9897±0.0018	0.8469±0.037	0.8584±0.028
Guo *et al*.[32]	0.9814±0.008	0.8305±0.006	0.9654±0.008
Sun *et al*.[33]	0.9780±0.010	0.8190±0.008	0.9650±0.009
Amin *et al*.[34]	0.7227±0.000	0.8070±0.000	NA
Zhao *et al*.[35]	0.9500±0.000	0.8600±0.000	NA
Zhang *et al*.[36]	0.9000±0.000	NA	0.8602±0.000
Proposed	0.9913±0.0004	0.8803±0.020	0.9911±0.004

NA: Not reported, IOU: Intersection over union, WT: Whole tumor

score, Dice coefficient, NETC-weighted dice, ETC-weighted dice, and mean pixel accuracy all improved by 17%, 2.4%, 11.2%, 14.8%, and 3.2%, respectively.

Comparison of different loss functions

For tumor segmentation tasks, ADAM is the best optimizer.[27] We compared the outcomes of the dice parameters as presented in Table 3 using ADAM as the optimizer and two alternative loss functions as described in section 3.1.1. In the investigation, the dice loss function outperformed the total loss function.

Comparison with existing work

We also compared the outcomes of our methods to those of other methods. Table 4 shows that the proposed method outperforms the other on every parameter. This could be due to two factors: first, the amount of data used to train the model, and second, the fact that the same model was trained on multiple datasets.

Conclusion

As we all know, due to the complexity of medical images, segmenting them is a difficult task. The proposed method effectively segments the tumor while requiring the least amount of time as compared to human experts. When compared to other state-of-the-art frameworks, this method had an accuracy of 99.0% or higher. The amount of time it takes to train the model and the amount of memory space it requires could both be reduced. Another extension of this would be to train the same model in different organs and tumor types (e.g., kidney, liver, and lung). We also want to convert this segmented tumor into a 3D Dicom format structure so that radiation therapy contouring can be done faster.

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Nil.

Conflicts of interest

There are no conflicts of interest.

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